

Business Case — BaSyx Editor Plugin Extension

Version History

Version	Date	Author	Comment
1.0	19.11.2025	Morten Haase	Initial structure / Prototype
1.1	30.03.2026	Martin Böhm	New author
1.2	08.04.2026	Martin Böhm	Rephrasing, structural improvements
1.3	12.04.2026	Martin Böhm	Updating costs

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Short Product Overview

The BaSyx Editor Plugin Extension adds several features to the existing BaSyx Editor, improving the handling of XML files in general and of KBL and VEC files in particular. It supports the upload, display, and processing of these files, and enables the creation of AAS content from their data.

Viability and Benefits

The BaSyx Editor Plugin Extension addresses a key need in the BaSyx ecosystem by streamlining the handling of XML-based engineering data. By enabling the upload, display, and processing of XML files — particularly KBL and VEC formats — it reduces manual effort and facilitates the creation of AAS content from structured external data. This improves usability for both technical users and non-experts, while enhancing interoperability and data consistency across systems. Since the extension builds on existing BaSyx components, it is technically feasible with low implementation risk.

Key benefits

- Reduced manual effort for XML file handling and AAS creation
- Improved usability for technical and non-technical users
- Enhanced data interoperability and consistency

Project Introduction

Stakeholders

Role	Organization / Detail
Customer Stakeholder	Rentschler & Wojcik
Provider	Team 3

Table 1 Stakeholders

Team / Roles

Role	Name	Email
Project Director	Martin Böhm	inf24132@lehre.dhbw-stuttgart.de
Product Manager	Florian Zahn	inf24254@lehre.dhbw-stuttgart.de
Test Manager	Daniel Ziegler	inf24202@lehre.dhbw-stuttgart.de
System Architect	Federico Dibenedetto / Felix Bandl	inf24072@lehre.dhbw-stuttgart.de / inf24227@lehre.dhbw-stuttgart.de
Technical Documentation	Leonardo Risatti	inf24074@lehre.dhbw-stuttgart.de

Table 2 Project team and roles

Cost Overview

Fixed Costs

Item	Cost (€)
Internet and Energy	1.000 €
Software Licences	700 €
Additional Hardware	200 €
Education	700 €
Total Fixed Costs	2.600 €

Table 3 Fixed costs breakdown

Variable Costs (Personnel)

Estimated personnel costs per 180 hours (as provided):

Role	Name	Hourly Wage (€)	Cost for 180 h (€)
Project Director	Martin Böhm	30	5.400
Project Manager	Florian Zahn	26,50	4.770
Test Manager	Daniel Ziegler	25	4.500
System Architect	Federico Dibenedetto	25	4.500
System Architect	Felix Bandl	25	4.500
Technical Documentation	Leonardo Risatti	25	4.500

Total (per 180 h): 28.170

Table 4 Variable personnel costs (per 180 h)

Costs per Work Phase

Phase	Martin Böhm	Florian Zahn	Daniel Ziegler	Federico Dibenedetto	Felix Bandl	Leonardo Risatti	Cost (€)
Analyse	40h	40h	20h	20h	20h	30h	4.510
Design	40h	40h	20h	40h	40h	40h	5.760
Coding	40h	40h	50h	60h	60h	50h	7.760
Testing	30h	40h	70h	40h	40h	50h	6.960
End Phase	30h	20h	20h	20h	20h	10h	3.180
Total	180h	180h	180h	180h	180h	180h	28.170

Table 5 Costs per work phase

Total Cost and Markup

The total project cost amounts to €30,770, comprising fixed costs of €2,600 and personnel costs of €28,170. After applying a 20% markup of €6,154, the final project price is €36,924.

Project Timeline

The Gantt chart illustrates the project timeline across five phases (Analysis, Design, Implementation, Testing, Finalization) with a total duration of approximately 34 weeks. Detailed planning is available in [Gantt project planner.xlsx](#).

ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE	PERIODS
						1234567891011121314151617181920212223242526272829303132333435
Overview:						
Analysis	1	6	1	8	100%	
Design	7	6	9	3	100%	
Implementing	13	20	13	18	90%	
Testing	20	13	19		0%	
Presentation	33	2				
Rollout	34	1				